Aaron N. Cutshall, DHA, MSHI

A Postgres Developer's Snowflake Survival Guide

SQLRV, LLC
HEALTHCARE INFORMATICS
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Doctor of Healthcare Administration

40 Years

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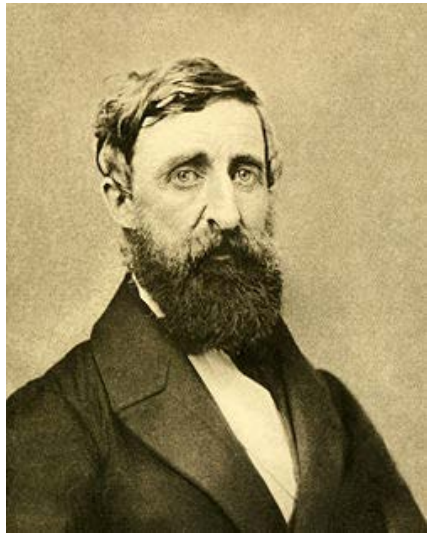
Something to consider…



credit

"Nature is full of genius, full of the divinity; so that not a snowflake escapes its fashioning hand."

– Henry David Thoreau (1817 – 1862)

A celebrated author of many science-fiction classics

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Agenda

- The History of Snowflake
- Snowflake Architecture
- Snowsight – the Snowflake IDE
- Programming features
- Review

2012-PRESENT

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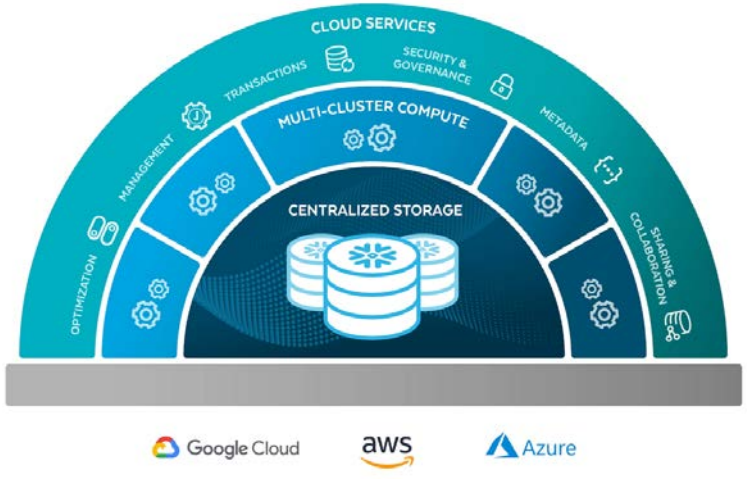
The History of Snowflake

- Created by Benoit Dageville, Thierry Cruanes, and Marcin Zukowski in 2012
- Most of the founding team enjoyed skiing and winter sports
- Each Snowflake is unique – one system overall yet shaped differently by each client
- Snowflakes are “born in the cloud”
- Cloud only data platform as a service
- Available with AWS, Google, or Azure



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Snowflake Architecture

- Fully managed service – No installations / updates to manage
- Cross-cloud consistency for replication / failover to another cloud region
- Few administrative settings to manage
- Security & regulatory requirements handled by Snowflake

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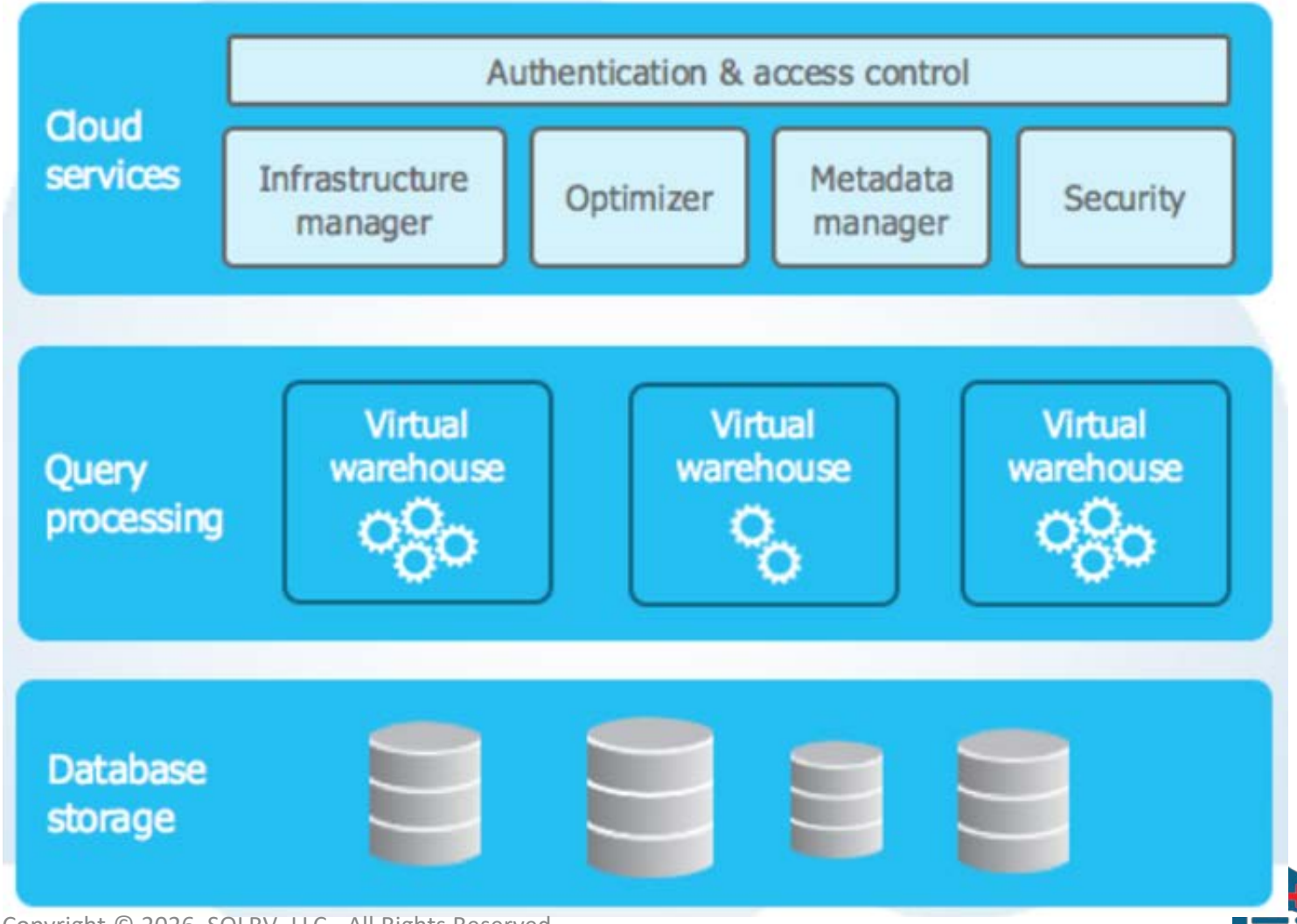
Snowflake Architecture

- Cloud Services
- Compute
- Storage



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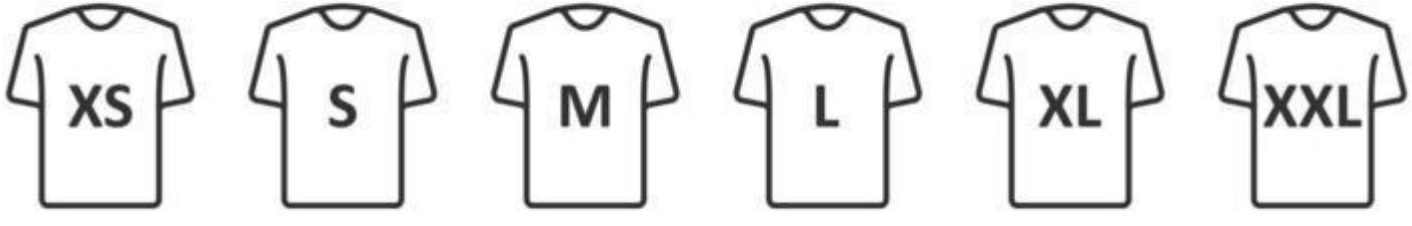
Snowflake Architecture

- Virtual Warehouse
  - How Snowflake organizes compute resources
  - Users must define a warehouse when connecting
  - T-shirt sizing to indicate compute resources: XS – 6XL
  - Easy to shift/assign appropriate resources per process
  - No need to scale up/down entire setup for all processes
  - Easier to manage costs on an as-needed basis



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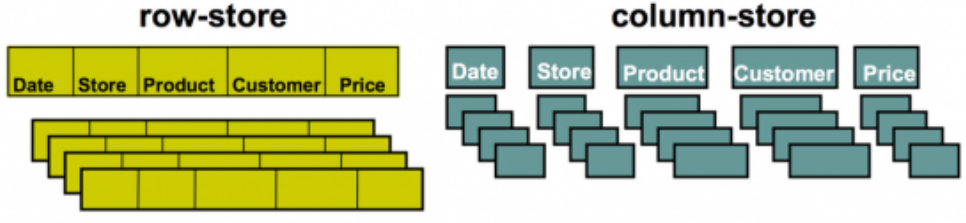
Snowflake Architecture

- Micro-Partitioning
  - Data divided into chunks of 50-500MB data (uncompressed)
  - Tracks value range, distinct values & other metadata
  - Range of values can overlap to reduce skew
  - Pruning is key for performance



- Columnar Storage
  - Columns stored independently within micro-partitions
  - Only columns referenced by a query are scanned
  - Automatic compression for each column

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Snowflake Architecture

Credit: docs.snowflake.com/en/user-guide/tables-clustering-micropartitions

Table: `t1`

Logical Structure

type	name	country	date
2	A	UK	11/2
4	C	SP	11/2
3	C	DE	11/2
2	B	DE	11/2
3	A	FR	11/2
2	C	SP	11/2
3	Z	DE	11/2
2	B	UK	11/2
4	C	NL	11/2
5	X	FR	11/3
1	A	NL	11/3
5	A	FR	11/3
2	X	FR	11/2
4	Z	NL	11/2
2	Y	SP	11/2
1	B	SP	11/3
5	X	DE	11/3
3	A	UK	11/4
1	C	FR	11/3
4	Z	NL	11/4
5	Y	SP	11/4
5	B	SP	11/5
3	X	DE	11/5
2	Z	UK	11/5

Physical Structure

	Micro-partition 1 (rows 1-6)			Micro-partition 2 (rows 7-12)			Micro-partition 3 (rows 13-18)			Micro-partition 4 (rows 19-24)		
type	2	4	3	3	2	4	2	4	2	1	4	5
	2	3	2	5	1	5	1	5	3	5	3	2
name	A	C	C	Z	B	C	X	Z	Y	C	Z	Y
	B	A	C	X	A	A	B	X	A	B	X	Z
country	UK	SP	DE	DE	UK	NL	FR	NL	SP	FR	NL	SP
	DE	FR	SP	FR	NL	FR	SP	DE	UK	SP	DE	UK
date	11/2	11/2	11/2	11/2	11/2	11/2	11/2	11/2	11/2	11/3	11/4	11/4
	11/2	11/2	11/2	11/3	11/3	11/3	11/3	11/3	11/4	11/5	11/5	11/5

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Snowflake Architecture

- Some design choices due to micro-partitioning
  - Metadata captured in same way as traditional databases
  - Few constraints are supported (NOT NULL and value range)
  - Such constraints are within a micro-partition
  - Constraints outside a micro-partition are not supported:
    - Unique Key (and Primary Keys that require uniqueness)
    - Foreign Key (referential integrity)
  - Choices allow other features that are more critical in a DW

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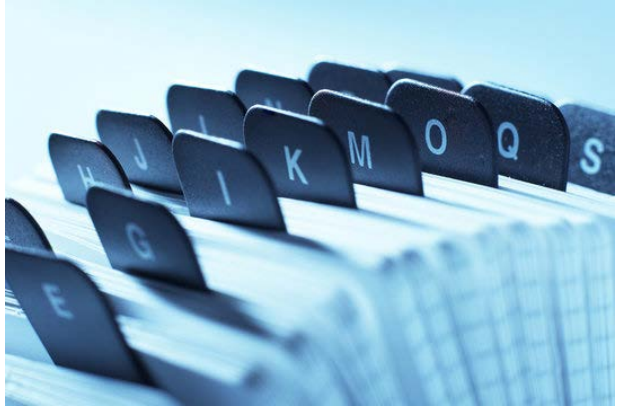
Snowflake Architecture



- Indexes
  - Indexes are not used in Snowflake
- Clustering keys
  - Used to sort rows within micro-partitions
  - Also used for micro-partition pruning for optimization
- Best results with:
  - Very large tables (>= 2 TB) and queried more than updated
  - Many query joins, filters, or sorts on the clustering key
  - High cardinality is critical (ex. DATE values)

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Snowflake Architecture

- Hybrid tables (new feature)
  - Similar to traditional transactional tables
  - Data stored at row level
  - Supports indexes & referential integrity constraints
  - Can be used along with standard Snowflake tables
  - Higher costs for maintenance on row storage clusters
  - Suggested for transactional or operational data store applications

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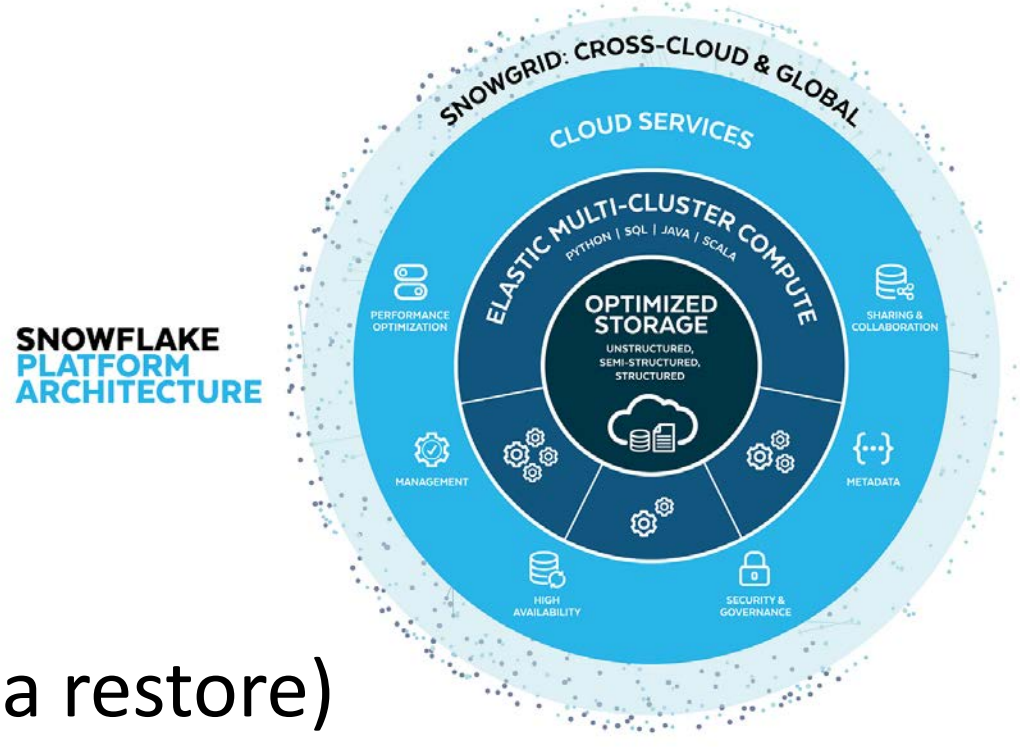
Snowflake Architecture

- Other capabilities
  - Zero-copy table cloning
  - Dynamic Data Masking
  - Row-level security
  - Secure data sharing
  - Semi-structured data
  - Time Travel (point-in-time data restore)
  - Iceberg tables (long term storage)
  - Dynamic tables (simplified Change Data Capture)



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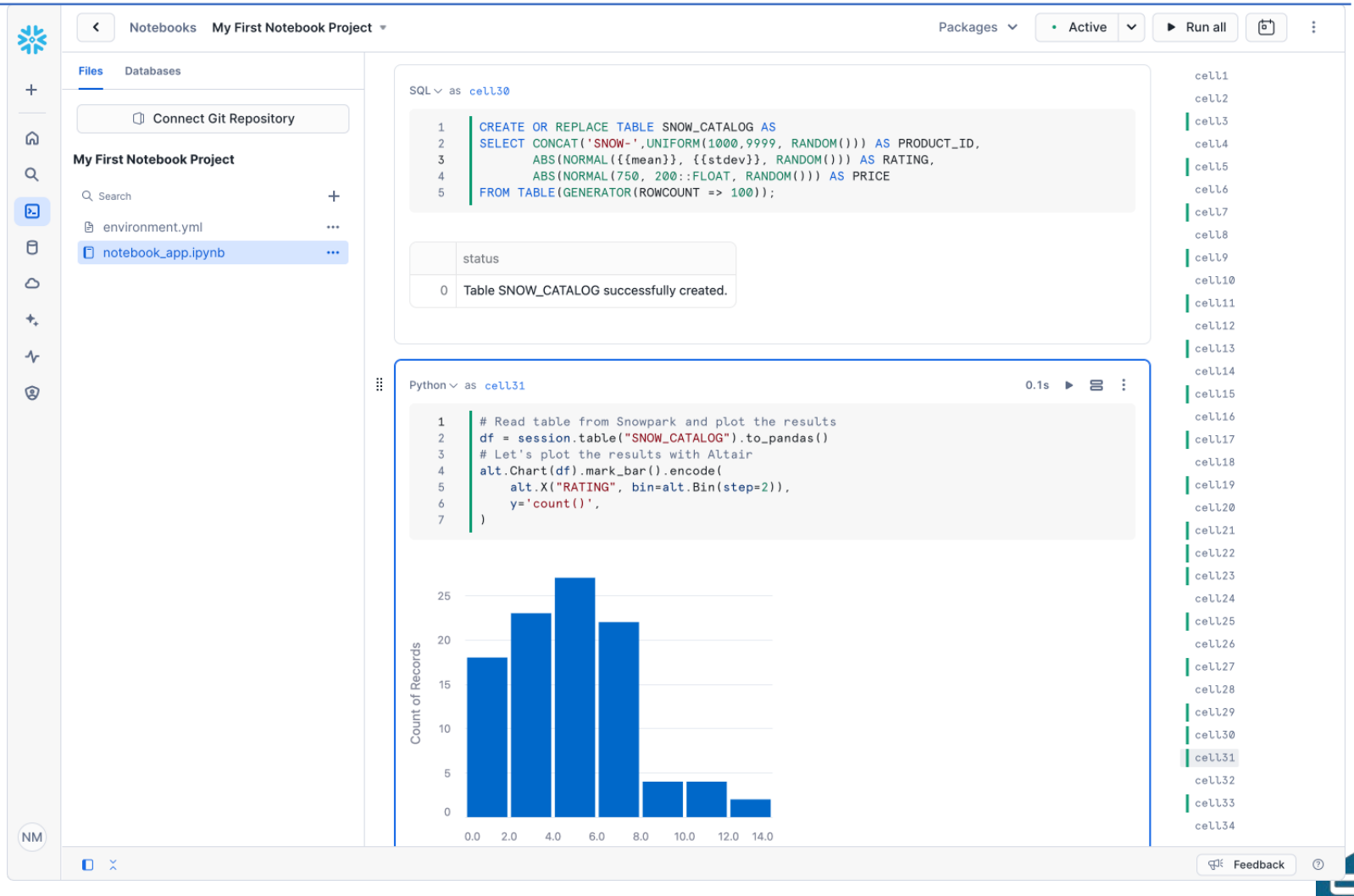
Snowsight: Snowflake IDE

- Web-based IDE further promotes “cloud first”
- Expected to be a one-stop shop
- Has everything needed for dev and admin
- Supported other IDEs if needed (ex. dBeaver)



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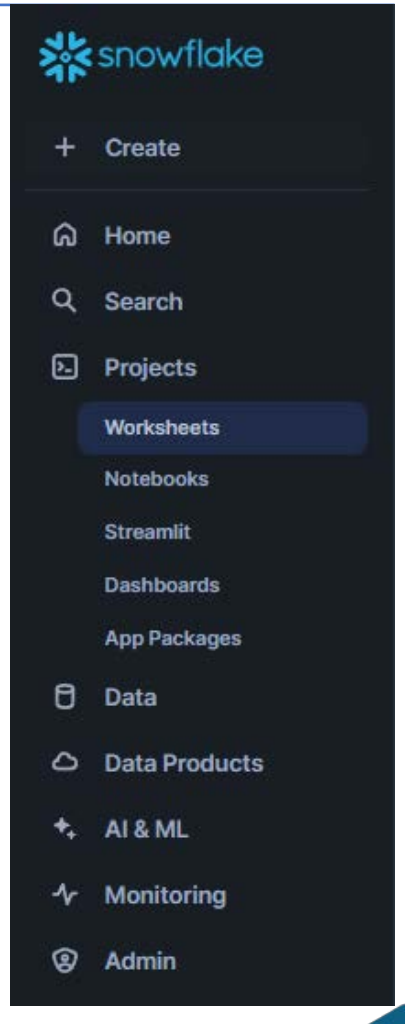
Snowsight: Snowflake IDE

- Projects
  - Worksheets – Primary SnowSQL interface for queries, views, stored procedures, & functions
  - Notebooks – Jupyter-style for SnowSQL or Python and can display graphs
  - Streamlit – Open-source Python library
  - Dashboards – Interactive tabular & graphical query results
  - App Packages – Create application containers for data, logic, metadata, and more



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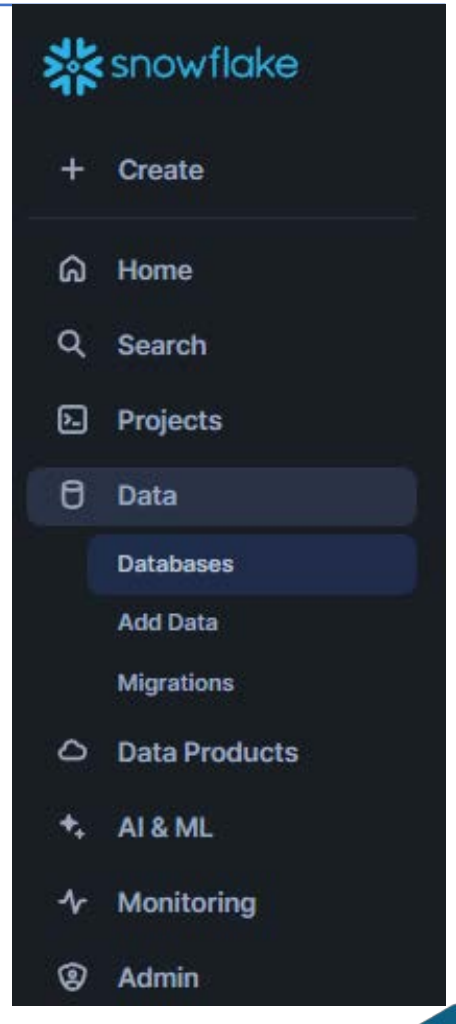
Snowsight: Snowflake IDE

- Data
  - Databases – Supports multiple databases per account including multiple layers of security
  - Add Data – Provisions to import data into Snowflake via the UI or other cloud-based connectors (AWS, GCP, Azure) & Postgres!
  - Migrations – Provides a wizard-like section to perform database migrations using SnowConvert and SnowPark



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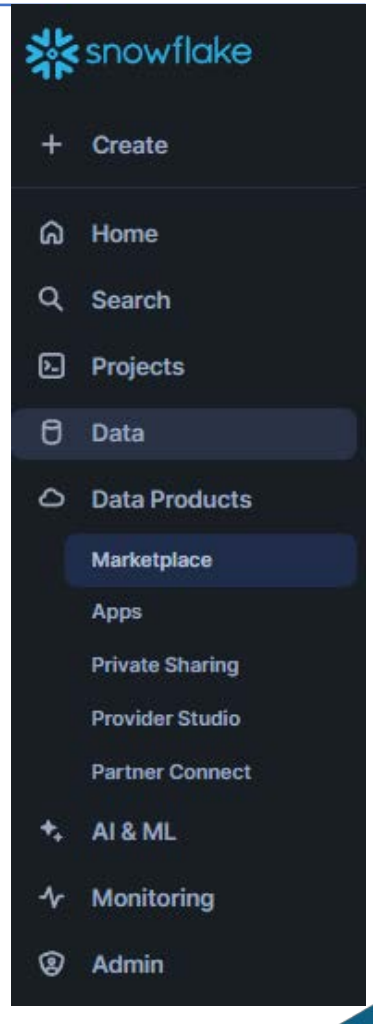
Snowsight: Snowflake IDE

- Data Products
  - Marketplace – A virtual market for applications, plug-ins, and other resources (many are free!)
  - Apps – Easy installer for apps or plug-ins from the marketplace or even an internal market
  - Private Sharing – Manage internal or external data sharing capabilities
  - Provider Studio – Manage published applications
  - Partner Connect – Manage business partner accounts



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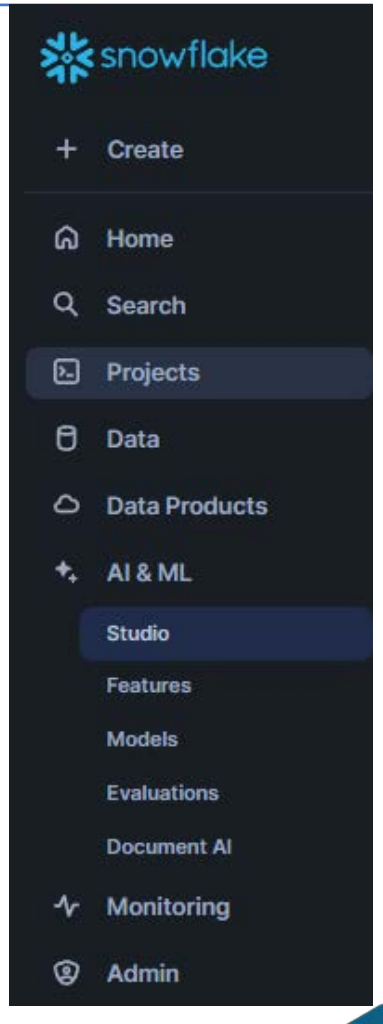
Snowsight: Snowflake IDE

- AI & ML
  - Studio – Various options to work with machine learning and large language models
  - Features – Lets data scientists and ML engineers work with ML features and store variations
  - Models – Stores ML models which can be shared
  - Evaluations – Manages LLM evaluations and results
  - Document AI – LLMs to extract data from unstructured documents



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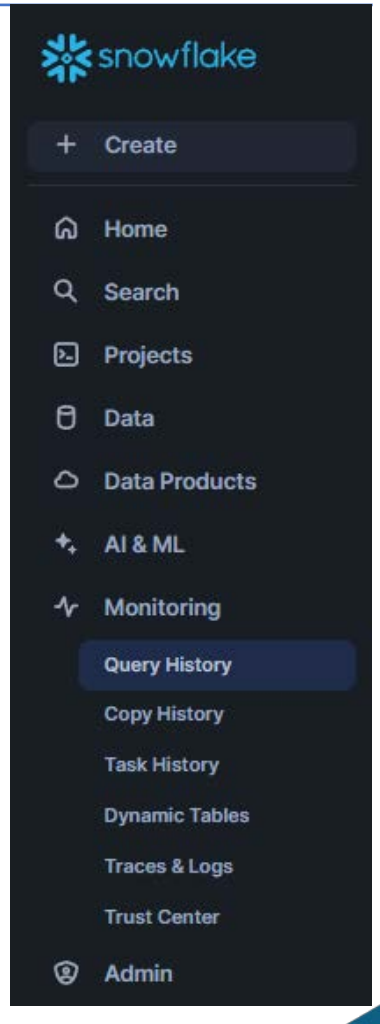
Snowsight: Snowflake IDE

- Monitoring
  - Query History – Dashboard of query history, statistics, and other queryable metadata
  - Copy History – History of data copied into and out of Snowflake
  - Task History – History of task execution & results
  - Dynamic Tables – Monitor dynamic table activity
  - Traces & Logs – Various debug methods
  - Trust Center – Monitor accounts and security



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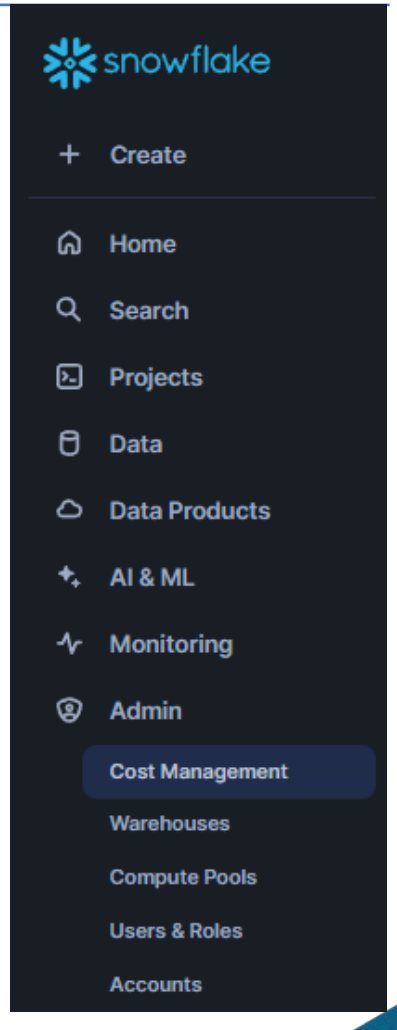
Snowsight: Snowflake IDE

- Admin
  - Cost Management – Various tools to help monitor costs and to set spend limitations
  - Warehouses – Manage custom resource configurations
  - Compute Pools – Collections of virtual machine nodes for Snowpark Container Services
  - Users & Roles – Manage all users and roles
  - Accounts – Fundamental Snowflake unit for managing all resources



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Snowflake Programming Features

- First Snowflake version only supported SQL for queries
- Later, supported logical views in SQL
- Now supports similar constructs:
  - Views – Logical and Materialized
  - Stored Procedures
  - Functions – Scalar and Tabular
- Multi-Lingual Procedures and Functions
  - Originally only in Javascript
  - SQL (procedural), Javascript, Python

```sql
SELECT DISTINCT
  c.customer_id,
  c.first_name,
  c.last_name,
  c.email,
  s.store_id
FROM
  rental r
  INNER JOIN staff st
    ON st.staff_id = r.staff_id
  INNER JOIN store s
    ON s.manager_staff_id = st.staff_id AND
       s.store_id = st.store_id
  INNER JOIN customer c
    ON r.customer_id = c.customer_id AND
       s.store_id = c.store_id
  INNER JOIN address a
    ON s.address_id = a.address_id
  INNER JOIN city ci
    ON ci.city_id = a.city_id
WHERE
  ci.city = 'Lethbridge';
```

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Snowflake Programming Features

- Borrowed features from other dialects
  - Data cast using :: notation (ex. my_integer_str::int)
  - Uses LIMIT *n* but also TOP *n* from T-SQL
- Similar support as in Postgres (not quite as good)
  - Arrays and array processing
  - Limited JSON functions & no native data type
- Similar or equivalent language functions to Postgres
- Most code can be translated without much difficulty

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Snowflake Programming Features

- No generate_series function!
  - Uses `TABLE(GENERATOR(ROWCOUNT => n))` functions to create rows
  - Uses `SEQ4()` function to create sequence

```
SELECT  SEQ4()+1 AS n
FROM    TABLE(GENERATOR(ROWCOUNT => 100)) v

SELECT  SEQ4()+2000 AS YEAR_NUMBER,
        DATE_FROM_PARTS(2000 + SEQ4(), 1, 1) AS YEAR_START_DATE,
        DATE_FROM_PARTS(2000 + SEQ4(), 12, 31) AS YEAR_END_DATE
FROM    TABLE(GENERATOR(ROWCOUNT => 100)) v
WHERE   SEQ4() < 100
```

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Snowflake Programming Features

- Cool feature: QUALIFY
  - Turns this:

```
WITH cte_LatestCustomPartOrder(CustomerID, OrderNbr, CustomPartNbr, RowNbr) AS (
          SELECT     oh.CustomerID, oh.OrderNbr, pd.PartNbr,
                     ROW_NUMBER() OVER (PARTITION BY oh.CustomerID ORDER BY oh.OrderDate DESC)
          FROM       Customer.OrderHeader oh
          INNER JOIN Customer.OrderDetail od
                     ON  od.OrderID = oh.OrderID
                     AND od.IsCustom = 1
                     AND od.IsDeleted = 0
          WHERE      oh.IsDeleted = 0
          )
SELECT    CustomerID, OrderNbr, CustomPartNbr
FROM      cte_LatestCustomPartOrder
WHERE     RowNbr = 1;
```

  - To this:

```
SELECT    oh.CustomerID, oh.OrderNbr, pd.PartNbr
FROM      Customer.OrderHeader oh
INNER JOIN Customer.OrderDetail od
          ON  od.OrderID = oh.OrderID
          AND od.IsCustom = 1
          AND od.IsDeleted = 0
WHERE     oh.IsDeleted = 0
QUALIFY   ROW_NUMBER() OVER (PARTITION BY oh.CustomerID ORDER BY oh.OrderDate DESC) = 1;
```

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Snowflake Review

- Scalability
  - Separation of compute from storage
  - Users can store an infinite amount of data
  - Scale compute resources up or down as needed
- Performance
  - Processes are separate to prevent interference
  - Resource contention prevented by separate process resources
- Cost-effectiveness
  - Storage and computation are paid for separately
  - Supports large data infrequent or simple queries
  - Also supports small data size with complex or frequent queries

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Snowflake Review

- Concurrency and accessibility
  - Multiple compute nodes
    - Simultaneously access the same data
    - No performance degradation
  - Supporting high levels of concurrency
- Flexibility and ease of use
  - Data distribution
  - Data partitioning
  - Query optimization
- Many more features not covered

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References

- Snowflake Documentation
- Snowflake vs Postgres: 5 Critical Differences
- From Postgres to Snowflake
- Snowflake for Postgres Lovers
- From PostgreSQL to Snowflake: A data migration story
- PostgreSQL vs Snowflake
- Snowflake Snowsight: 7 refreshing features

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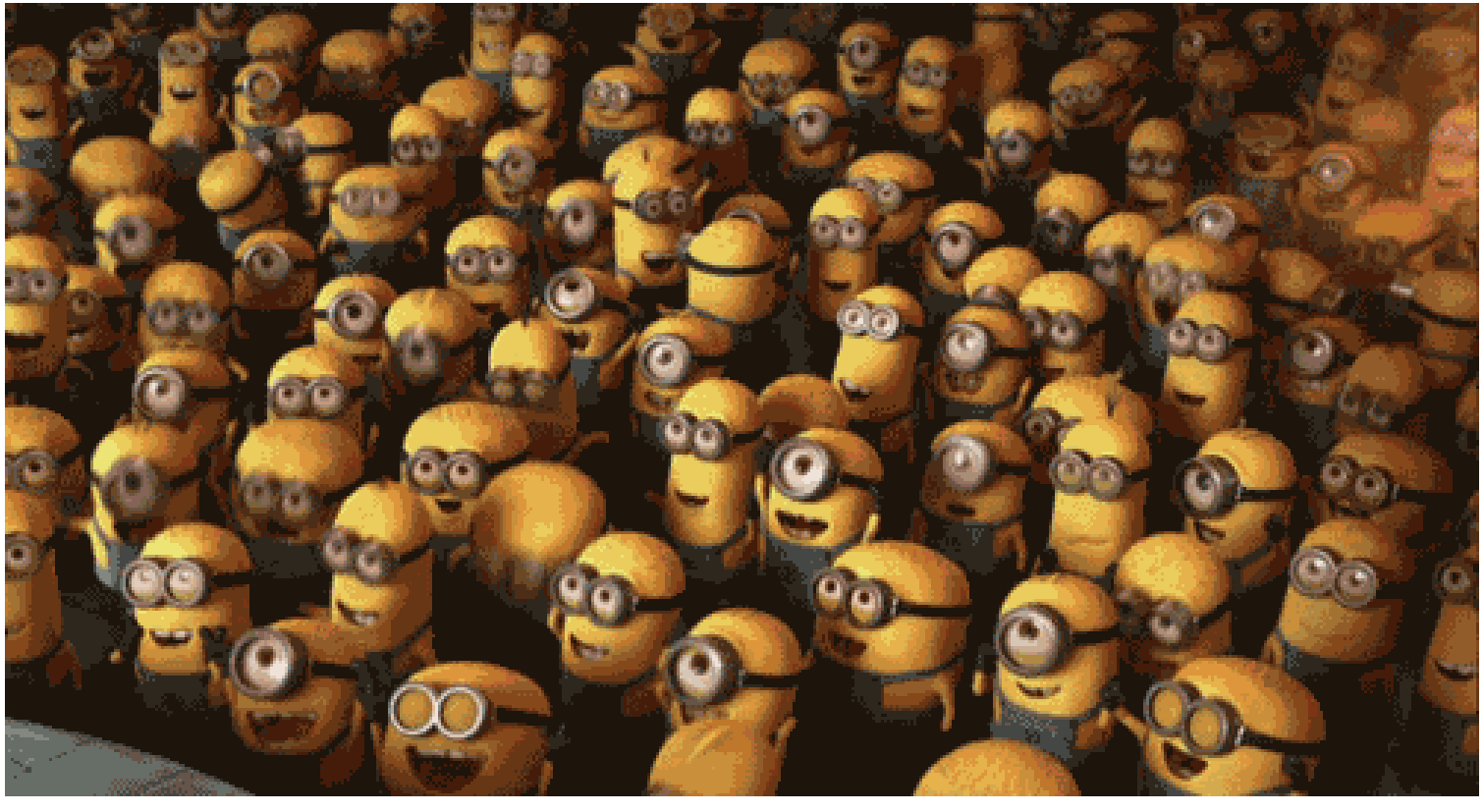

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